

Haptic Hongi – Reiterated

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ABSTRACT

Haptic Hongi – Reiterated uses Augmented Reality (AR) to recreate meeting an indigenous person on their own terms, and thereby reimagine first encounters that took place during the pre-colonisation era in Aotearoa New Zealand. These encounters were often asymmetrical affairs, freighted with troubled power dynamics and violence, so there is a need to rebalance, restore and reconnect in a playful way. In the experience, AR technology is used to place a virtual woman in front of the user, and then haptic actuators are used to enable the user to feel a hongī, the traditional Māori greeting involving pressing of noses. In this way, AR technologies enable people to communicate across time and space so that they may feel as though they have met.

CCS CONCEPTS

• Applied computing → Media arts; • Computing methodologies → Mixed / augmented reality.

KEYWORDS

Intercultural, Indigenous, Māori, Haptic

ACM Reference Format:

Mairi Gunn, Tania Remana (Ngāpuhi), Prasanth Sasikumar, Sachith Muthukumarana, and Mark Billingham. 2023. Haptic Hongi – Reiterated. In *SIGGRAPH Asia 2023 XR (SA XR '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610549.3614618>

1 INTRODUCTION

This project explores how Augmented Reality (AR) and haptic technology can recreate a first encounter between a Māori indigenous person and non-Māori in pre-colonial Aotearoa New Zealand. It is an expanded version of the AR installation *Come to the Table! Haere Mai ki te Tēpu!* [Gunn et al. 2019]. The main extension is to add a hongī, a pressing of noses and foreheads, a sharing of breath, a traditional Māori greeting. Our research explores how new and



Figure 1: The sponge-coated actuator presses the user's nose

emerging Extended Reality (XR) technologies might provide an opportunity to investigate and trial their potential to reverse processes of atomisation, polarization, intercultural discomfort and mistrust in our contemporary society. This practice-led, transdisciplinary research was inspired by community economic thinking, particularly a commoning solution to current dilemmas that confront and challenge us. The collaboration between former strangers from various disciplines and the broader community allowed us to discuss our current societal stressors, prioritise relationality and explore our shared values while we created XR experiences as activators to stimulate dialectical discourse and more. For a longer discussion about the research motivation, see [Gunn et al. 2021].

2 THE TECHNOLOGY

This work has three key elements: (1) Volumetric capture, (2) AR display, and (3) haptic feedback. In this section, we describe each.

2.1 Volumetric capture

The virtual video of Tania is shown as a 3D volumetric video. We used the DepthKit software [DepthKit 2023] for recording the volumetric capture. This was chosen due to its lightweight encoding that enabled us to have a completely stand-alone and portable solution.

This software currently only supports single-camera input, which is one of the limitations of our system. However, this suited our needs as the encoding used is easy on computing resources and, due to the table-top setup, all users are seated opposite the volumetric character who is always gazing directly towards the user. In

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SA XR '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0316-4/23/12.

<https://doi.org/10.1145/3610549.3614618>



Figure 2: Tania captured as a volumetric video

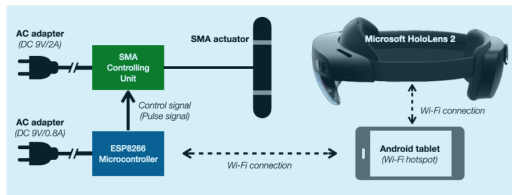


Figure 3: High-level system over-view

this case we used the Azure Kinect depth sensing camera [Azure 2023]. One of the primary technical hurdles we encountered was recording and playing back 3D volumetric content. Specifically, we grappled with managing the file size and ensuring smooth playback performance. To address these challenges, we conducted experiments using compression and playback algorithms, including Intel Realsense SDK and Azure Kinect SDK. After thorough testing, we ultimately opted for DepthKit, as it struck the right balance between compression and performance and boasted a frame rate of 30 frames per second.

2.2 AR Display

We wanted to create the illusion that Tania was really at the table with the user, speaking directly to them. To achieve this, we viewed the volumetric data on a HoloLens 2 AR head-mounted display [Microsoft 2023]. The HoloLens 2 is an optical see-through head-mounted display with a 43-degree diagonal field of view and a resolution of 2048×1080 pixels per eye. The HoloLens 2 also has a number of cameras on that display that can perform SLAM-based tracking, ensuring that the AR content appears fixed in the real world. When wearing the display, the user sees the virtual video of Tania as a solid image fixed in space, creating the strong impression that she is really sitting across the table from them.

2.3 Haptic Feedback

The *haptic HONGI* framework was built to support peripheral devices for haptic feedback. The haptic feedback unit was built as a complete standalone module that would respond to commands

from the main application running in the AR display. The haptic feedback unit had two power supplies - one for the microcontroller and one for the SMA actuator and its controlling unit. We used an ESP8266 Wi-Fi microcontroller due to its built-in TCP/IP networking capabilities. We used an Android tablet as a portable Wi-Fi hotspot to which both the microcontroller and the AR display would be connected. The actuator uses a Shape Memory Alloy (SMA) strip from Toki Corporation (JP), a bimetallic material that can change length through applied electrical current or heat. We designed a 3D-printed housing to accommodate an SMA loop. The SMA wire strip can warm up to 60°C when activated. We attached small, sculpted pieces of makeup sponge to both moving ends to realise gentle haptic feedback and to prevent direct contact of the SMA with the visitors' facial skin. Once all devices are in operation, a command to activate the actuator, issued by the AR application, is picked up by the microcontroller (connected to the same network via Wi-Fi hotspot). The microcontroller would then activate the SMA actuator, moving the makeup sponge against the nose and forehead to create the feeling of sharing a hongi.

3 CONCLUSION AND NEXT STEPS

In this paper, we have described the Haptic Hongi system that uses AR and haptic technology to represent the traditional Māori greeting that includes a hongi. This is part of a broader effort to address intercultural discomfort in society. The feedback has been very positive, with visitors finding the experience informative, engaging and even emotionally moving. Visitors from migrant backgrounds seemed particularly relieved when Tania greeted them so warmly. Māori visitors surprisingly noted that they felt 'the wairua', her spirit, cross time and space. Pressure from the actuator accentuated their feelings. However, there are areas for future research that could improve the system. We would like to explore eye-tracking to enable the virtual character to appear to make eye contact. We will transfer the experience to a Magic Leap 2 that has a taller field of view and a unique feature that performs segmented dimming, which makes the AR content appear solid. The research process itself brought diverse researchers together in positive and constructive ways. This demonstrated the potential to overcome siloed thinking and competitive practices to create a compelling interactive piece.

ACKNOWLEDGMENTS

A University of Auckland Doctoral Scholarship and a Ria McBride Research Award from Hine Kahukura New Horizons for Women supported this research.

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